Korean Air cargo Flight 8509

Flight model: Boeing 747
Date: December 22, 1999

From: Uzbekistan
To: Italy

The plane starts from South Korea to to Italy with a stopover at Uzbekistan and UK. With, a crew change at UK to Italy. The captain is one of the most respected pilot in the airline.

Problem

There is a delay in take off as the ground staffs didn't file a **flight plan** yet, A flight plan is an mandatory filing which is to be done prior to take off. After which the plane takes off. Shortly after take off the pilots starts to turn left but, the captains **Artificial horizon instrument** says the plane isnt turning. The flight engineer warns the pilot that the plane is banking but the pilot's seems to not take any action for the same. Then the plane crashes.

Investigation

The airport authorities had informed the investigators that the runway was covered with debris which suggests that the plane actually began to disintegrate on take off. But, in actual these debris were carried by the wind from the crash site. The investigators concern that uranium depletor over potential radioactive hazard, uranium depletor is used to improve stability in flight. TO rule out the possibility of in-flight brake up the investigators check the "4 corner search" which include 2 wings, tail plane, fine and rudder then it means the flight us intact right before crash, which was found at the crash site. There was deep marks on the ground which was caused due to the plane wing as it was banking banking left at the time of crash. Soon it was found that the captains ADI (Artificial directional indicator)/ Artificial horizon had failed to work properly. The airline should have maintained 2 maintenance record on in flight and other in the ground. But, the company only had one record which was destroyed in flight.Upon interview with the previous crew it was found that the ADI problem was brought to the maintenance crew's notice.They did believe that it was a problem with the connectors connecting the instrument and had replace the connector which was not the actual fault. Upon looking into the data recorder it was found that the plane roll never exceeded 2 degree which was faulty. The 2 **INU (Intial naviagtion unit)** feed roll and pitch data to 3 of the pilot's ADI (1 captain, 1 first officer and 1 standby). which had delivered faulty reading to the captain. The DVR shows that there was a warning horn when the engines were turned on which the crew ignored, The alarm which popped up tells the pilots one of their ADI's are malfunctioning and must check with the standby horizon. The flight engineer did point out the defect to the captain but the captain didnt respond and first officer didn't say anything despite having working instrument. The captain sounded more agitated right before take off and was more rude with the crew. The pilots could have brought back the plane if they had responded to the warnings of the flight engineer. The flight training of the Korea air shows that there was a hierarchy between the first officer and the captain and was considered as the captain to be more wiser and experienced and had no free thinking. The captain who worked as a pilot in fighter jet (Sole) had no/ little experience with flying in commercial jets. The crews in general are to trust the instruments and not senses.

Changes that were brought after the incident

The airline was bought under strict scrutiny around the world. The investigation board recommended the airlines in Korea to modify the training with the view of Korean culture which emphasize that the elders are more wise and experienced that the younger and must obey their orders. All these improved their training program.

Reference

1. London Stansted Airport Crash — Mayday: Air Disaster
   https://www.youtube.com/watch?v=RoKtlv5KqQo